CryspLight full simulation: working plan

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1 Scope

Build the full simulation specified in `fullsim_design.tex` and compare it with the bench measurements: BGO and CsI(Tl) at ambient temperature with the DT2745 digitizer first and then the Citiroc 1A (data taking the week of 20 October 2026), BGO at 196 K with the digitizer afterwards, Radioroc2 when the board arrives. The first study is the characterisation of BGO at ambient temperature (Section 3). This document tracks what is planned, in progress and done, the decisions taken and the questions open. It is edited at every step; the physics stays in the design document.

2 Principles

- Analysis in Julia (`fullsim_analysis/lib`, `scripts`), outputs as HDF5 and CSV under `fullsim_analysis/results/`, figures by Python scripts under `fullsim_analysis/plots/` from those files.
- The pedagogical simulation is the git tag `pedsim` and the branch `main`; the full simulation grows on the branch `fullsim`. Nothing is replaced: every feature is gated by a configuration key and the frozen-output test proves the pedagogical path unchanged.
- Repository layout for the full simulation: `docs_fullsim/` (documents), `fullsim_runs/<campaign>/` (simulation and analysis configurations plus the run log of each measurement campaign), `fullsim_analysis/` (Python comparison layer, committed figures under `results/<campaign>/`), engine additions in `src/`, the electronics emulation in `src/frontend/`, tests in `test/`. Simulation output in the gitignored `output/<tag>/`. Measured data outside the repository, referenced by path from the campaign configuration.
- Documents are written before the code they describe; results are produced by tracked scripts and committed with their images; every reported result states where it lives.

3 Phases and tasks

Phase 0 — before and during the ambient runs

id	task	status	date
0.1	Tag `pedsim`, branch `fullsim`, datasheets committed	done	16 Oct
0.2	Design document and this plan	done	16 Oct
0.3	Overvoltage recipe for the bench (`BGOTSCAN/docs/overvoltage_recipe.tex`)	done	16 Oct

id	task	status	date
0.4	Frozen-output test of the pedagogical path: SHA-256 digests of three CPU runs (point source through the reference transport and through the kernel, gamma pipeline) in `test/frozen_checksums.toml`, compared by `Pkg.test`; regenerated only by `test/frozen.jl -freeze`	done	17 Oct
0.5	Readout layout: tiled array (active squares, margins, seam, overhang) in the `Readout` of both transport versions, dead area as a reflective or absorbing surround; layout, conservation and parity tests	done	17 Oct
0.6	Grease and resin window: second Fresnel interface with the round trips in the thin coupling layer (`n_window` in `OpticalParams`, 0 = none); analytic, light-loss and parity tests	done	17 Oct
0.7	Materials: `bgo_cold` (2720 ns, 15 000 ph/MeV midpoint), `bgo_ambient` (300 + 60 ns, 7600 ph/MeV midpoint), `csitl_ambient` (730/3340 ns, 64/36%); two-component emission in both transport versions and in the event pipeline; tests	done	17 Oct
0.8	^{22}Na point source: whole decays (1275 keV always, 511 keV pair in 90.3%), entry of each gamma on the crystal, flight time, per-decay pipeline `run_decays!` with interaction lists, photoelectron records and maps, chunkable by decay range; acceptance and pipeline tests	done	17 Oct
0.9	Photoelectron file `output/<tag>/pes.h5` (groups events, interactions, pes with offsets; configuration and code version in the attributes), written chunk by chunk; configuration reader for `fullsim_runs/` and the driver `scripts/fullsim_optical.jl`; first configurations of the two campaigns; round-trip, re-binning and driver tests	done	17 Oct
0.10	SiPM device model `src/frontend/sipm.jl`: breakdown, gain, PDE (overvoltage and wavelength), crosstalk, afterpulsing with recharge, dark counts, pulse shape, per-channel gain spread; the records are thinned by the PDE and turned into avalanche trains; device parameters in `data/sipm_s14160.toml`; unit tests	done	17 Oct
0.11	Citiroc chain `src/frontend/citiroc.jl` on the grid of `shaping.jl` (first-order sections, no transform): slow and fast semi-Gaussian shapers calibrated to one avalanche, shaped noise, discriminators and majority trigger, peak detector or track-and-hold, ADC with pedestal, input-charge and ADC saturation, low-gain path; unit tests, the stored fractions 0.39/0.15/0.07 reproduced	done	17 Oct
0.11b	Digitizer chain `src/frontend/digitizer.jl` on the avalanche train: gate and baseline weights from the pulse integral, integrated white noise, sampled-amplitude trigger with majority, saturation; unit tests (window fractions, baseline subtraction, thresholds, noise rms, gate length). Waveform subsample not yet written	done	17 Oct

id	task	status	date
0.12	Front-end stage `src/frontend/frontend_run.jl` and `scripts/fullsim_frontend.jl`: SiPM response per channel and one front end per decay, multithreaded and reproducible per event, writing `events.h5` (/run, /channels, /calib, /events, /mc); four front-end configurations of the ambient campaign; first-look `fullsim_analysis/summary.jl` and `trigger_scan.jl`. The converter of the bench files is JJ's	done	17 Oct
0.13	First emulated campaign at the bench settings (table below), the summaries of the seven event files, the trigger scans (ambient BGO with the three front ends, cold BGO with digitizer and Radioroc2), the shaping scans of the Radioroc2, the gate scans of the digitizer and the dynamic-range studies, all as tracked scripts with outputs under `fullsim_analysis/results/`. Still to do: the overvoltage scan (2 to 5 V) and the trigger scan for CsI(Tl)	in progress	18 Oct
0.14	Settings and procedure delivered to the bench: `BGOTSCAN/docs/digitiser_run_instructions_and_procedure.tex` (version 1, 18 Oct; the gate start and the option-A numbers to be filled from the scans)	in progress	18 Oct
0.16	Analysis layer: Julia library `fullsim_analysis/lib`, one script per study under `scripts/` writing HDF5 and CSV, Python plotters under `plots/`	done	17 Oct
0.17	Position resolution on the emulated maps of the ambient BGO study, photopeak and single interaction, with the Anger and CNN reconstructions of the pedagogical simulation reading the emulated charge maps; one number per front end (moved from 2.5)	planned	
0.15	Radioroc2 chain `src/frontend/radioroc.jl` (per-channel shaping, noise falling with the peaking time, T1/T2 on the time preamplifier and TQ on the low-gain shaper, hold, both paths with their ceilings) on the branch `radioroc`; configurations of the three crystals; shaping-time scan `fullsim_analysis/shaping_scan.jl`; the cold comparison Radioroc2 against digitizer	in progress	17 Oct

Phase 1 — when the ambient data exist

id	task	status	date
1.1	Converters from the bench files to the common layout, with the run log: Citiroc charges; digitizer waveforms reduced to baseline, gated charge and threshold crossings (from the Antarctica analysis code)	planned	
1.2	Single-photoelectron gains and breakdown per channel; pedestals and noise term; dark rates	planned	
1.3	Spectra: 511, 1275, sum peak; resolution and its decomposition; 1275 to 511 ratio	planned	

id	task	status	date
1.4	Light yield in photoelectrons and in photons per MeV through the simulated collection	planned	
1.5	Maps: occupancy, mean charge, seed-channel photopeak; fit of the dead-area reflectivity and check of the roughness	planned	
1.6	Brightest-channel distribution and saturation in CsI(Tl)	planned	
1.7	Emulation rerun with the measured noise, gains and settings; side-by-side figures	planned	
1.8	Note section per campaign in the results document	planned	

Phase 2 — accuracy and the next setups

id	task	status	date
2.1	Spectral sampling: emission spectra, $n(\lambda)$, PDE(λ); spectral test	planned	
2.2	Cold BGO campaign at 196 K: breakdown measured in situ, emulation, comparison; warm-to-cold yield ratio against the quenching curve	planned	
2.3	Radioroc2 campaign on the bench (the chain is task 0.15)	planned	
2.4	Gamma stage: K-shell fluorescence and escape	planned	
2.5	Position reconstruction on emulated maps for the other configurations (the ambient BGO case is task 0.17)	planned	
2.6	Design document version 1.0 with the measured parameters	planned	

The first study: BGO at ambient temperature

Agreed on 18 October. The complete characterisation of the BGO crystal at room temperature with the three front ends, the digitizer first because it is on the bench with the Citiroc and gives the cleanest energy measurement, then the Citiroc, then the Radioroc2 when it arrives, and the comparison between them. The questions, each answered by the emulation now and by the data when they exist:

1. Whether the system can be calibrated with dark pulses and one gain: single-photoelectron amplitude and breakdown per channel (the Citiroc high-gain path, the digitizer at gain 4, the Radioroc2 high-gain path).

2. The energy resolution at 511 keV, with the noise and statistical terms separated from the intrinsic one.

3. The light efficiency, from photons emitted to photoelectrons: the collection of the optics and the detection efficiency, and the fraction of the photoelectrons that each front end stores.

4. The charge per SiPM and the charge of the seed channel, their distributions and the fraction of events in which the seed exceeds the dynamic range of the front end.

5. The position resolution for photopeak events with a single interaction, which depends on the light left in the dim channels after each front end, and on the efficiency of that selection.

For the Radioroc2 the agreed operation is the default gains of the chip (high gain 5.3, low gain 0.53), both paths read: the high gain for the calibration with dark pulses and for the dim

channels, the low gain for the energy of every channel, cross-calibrated against the high gain on the channels both see between 10 and 70 avalanches; the longest shaping, 1.8 µs; the charge trigger TQ as the acquisition trigger.

First emulated campaign (17 October 2026)

Optical runs with the intrinsic resolution of 5% FWHM at 511 keV; front ends at the bench settings: Citiroc at 3.0 V (BGO) and 2.7 V (CsI(Tl)), 87.5 ns shaping, peak detector, hold 300/500 ns, single-channel threshold of 8 photoelectron-equivalents; DT2745 at 43 V and gain 1, threshold 120 counts with majority 2, gate 1 µs (BGO) and 4.5 µs (CsI(Tl)). Energies in avalanches of the stored or integrated signal, summed over the 64 channels.

	BGO Citiroc	BGO DT2745	CsI(Tl) Citiroc	CsI(Tl) DT2745
decays simulated	400 000	400 000	100 000	100 000
triggered fraction of decays	19.5%	19.4%	14.6%	14.8%
511 keV peak (avalanches)	833	1973	1635	10 867
FWHM at 511 keV	8.8%	8.9%	6.6%	6.3%
1275 / 511 peak ratio	2.31	2.49	2.34	2.50
brightest channel, median / 1%	30 / 103	74 / 274	43 / 261	272 / 1949
saturated events	0	0	0.74%	0

Three remarks. The Citiroc stores 0.42 of the digitizer's photoelectrons for BGO and 0.15 for CsI(Tl), the fractions of the design document once the ratio of the detection efficiencies at 3.0 and 5 V is taken out. The 1275 to 511 ratio of the Citiroc is 2.31 instead of 2.5 because each of the 64 peak detectors stores the maximum of its noise and dark pulses when the channel receives little light, a pedestal of about 120 avalanches per event that the digitizer's baseline subtraction removes; the measured ratio will show the same and the per-channel pedestal of empty events corrects it. The CsI(Tl) Citiroc saturates in 0.7% of the events, the brightest channel of the events next to the readout face, as predicted.

4 Campaigns

campaign	configuration	readout
`2026-10_ambient`	BGO ambient, CsI(Tl) ambient	DT2745 first, then Citiroc 1A ×2, on the sam
`2026-xx_digitizer_cold`	BGO 196 K	DT2745
`2026-xx_radioroc`	all three	Radioroc2

5 Decisions

16 Oct 2026 Full simulation as a subproject of CryspLight, not a client package: most of the work is engine work under the parity test; one history, one test suite.

16 Oct 2026 Additive only. Tag `pedsim`; frozen-output test; gated features; separate files for the photoelectron records and the emulated events; pedagogical configurations and outputs untouched.

16 Oct 2026 Three-stage pipeline with files between stages, so that the front end can be re-tuned against data without rerunning the optics.

16 Oct 2026 Grease ($n = 1.42$) in every configuration; not a parameter.

16 Oct 2026 Cold BGO decay time 2.7 µs from the Antarctica runs, in a new material file; the Ding file stays for the pedagogical runs.

16 Oct 2026 The Antarctica runs are not simulated (dead channels, noisy chain); the first comparison is the Citiroc ambient campaign.

16 Oct 2026 Overvoltage set from the measured breakdown, 3.0 V for BGO and 2.7 V for CsI(Tl); recipe in `BGOTSCAN/docs/overvoltage_recipe.tex`.

16 Oct 2026 Measured and emulated events share one HDF5 layout; the bench files are converted to it as the first analysis step.

16 Oct 2026 No waveform-digitiser emulation. *Reversed the same day*, see below.

16 Oct 2026 The DT2745 digitizer is a third front end of the simulation, emulated on the avalanche train (gated charge with baseline subtraction, integrated noise, sampled-amplitude trigger) rather than through waveforms. Reason: the Citiroc at 87.5 ns stores 0.07 of the photoelectrons of cold BGO, so the cold runs are taken with the digitizer, and the ambient runs with both readouts give a parameter-free test of the stored fraction. The waveform processing of the measured data (baseline, single-photoelectron search) stays on the data side.

16 Oct 2026 Digitizer noise: nominal (4 counts rms, white, independent channels) until the new runs; the low-frequency and group-correlated terms are added then as a measured input.

16 Oct 2026 Afterpulsing 1% at 3 V, scaling with the square of the overvoltage, 100 ns; not resolvable with the Citiroc.

16 Oct 2026 Intrinsic resolution 5% FWHM at 511 keV for both crystals; the non-proportionality curve of BGO (Gentile et al.) as a later refinement; Moszynski et al. to be checked. *17 Oct:* implemented as one Gaussian factor on the light of each decay, of relative width $0.05/2.355 \times (511\ \mathrm{keV}/E)^{1/2}$; the exponent is an assumption to be checked on the width of the 1275 keV line (keys `intrinsic_fwhm_511`, `intrinsic_exponent` of the material files).

16 Oct 2026 Ambient digitizer runs at gain 1 and 43 V (5 V of overvoltage), the simplest setting for the bench; record 10 µs with 1 µs pre-trigger. Threshold 60 counts, majority 2, *corrected on 17 Oct* by the emulated scan to 120 counts with majority 2 (or 160 single channel); Citiroc at 3 V: 8 photoelectron-equivalents single channel (or 5 with majority 2). Recipe and configurations updated.

6 Open questions

- Reflectivity of the tile surface between channels (default diffuse 0.5; to fit on the occupancy maps).
- BGO light yield at ambient and at 196 K for this crystal; BGO intrinsic resolution.
- Citiroc settings of the ambient runs (shaping time, gains, hold delay, thresholds, trigger definition in the FPGA) and the ADC; to be written in the run log.
- Angular dependence of the PDE through the resin from grease.
- Citiroc shaper transfer function and noise spectrum; whether a single-photoelectron waveform can be probed on the test board.
- Temperature of the SiPMs in the cold run (resolved by the measured breakdown; a thermometer on the SiPM board would still help).

- Whether the digitizer gain stage lowers the noise per photoelectron (baseline runs at gains 1, 2, 4, 8 with the SiPMs unbiased); decides the overvoltage of the cold runs.
- Whether the Radioroc2 at 1.8 µs shaping (0.55 of the photoelectrons of cold BGO) is usable in the cold, or a less cold point is preferable. First emulation (18 Oct): 7.2% FWHM at 511 keV against 6.7% for the digitizer with a 10 µs gate; usable, the digitizer better.
- The low-frequency and group-correlated noise of the digitizer on the bench cabling, measured in step 1 of the procedure; the emulation assumes white noise.
- The acquisition board and ADC of the Radioroc2 (assumed: the evaluation board, 12 bits, multiplexed readout) and whether its FPGA offers a majority; the charge trigger TQ needs neither.
- Digitizer gate start before the trigger: the gate scan running on 18 Oct decides between 200 and 300 ns; the instructions say 300 ns.
- The format of the WaveDump2 output files and the event-building of the 64 channels, for the converter.

7 Log

16 Oct 2026 Project assessed; datasheets read (Citiroc 1A, Radioroc2, S14160/S14161); Antarctica draft read; tag and branch; design document v0.1 and this plan; overvoltage recipe written.

18 Oct 2026 Decisions on the Radioroc2 operation (default gains, both paths, energy from the low gain, calibration from the high gain, longest shaping, charge trigger) after the dynamic-range study showed the high-gain ceiling of 72 avalanches exceeded in 14 to 47% of the 511 keV events and the register ranges not allowing a higher one. The digitizer is the first front end of the study; its settings and procedure for the bench written as `digitiser_run_instructions_and_procedure.tex` with two options (gain 4 at 3 V of overvoltage if the noise allows, else gain 1 at 5 V), record 8 µs with 1.5 µs pre-trigger, threshold 7.5 photoelectrons with majority 2, gate from 300 ns before the trigger. The Antarctica runs are not a reference: only the single-photoelectron amplitude per volt and the pulse shape are taken from them, as expectations that the procedure measures.

16 Oct 2026 Design document reviewed with JJ: measured data (one high-gain charge and the hit register), afterpulsing, the stored fraction recomputed with the avalanche pulse (0.39/0.15/0.07), digitizer manual (UM8717) read and the DT2745 added as a third front end; digitizer section added to the overvoltage recipe; trigger thresholds for the Citiroc and the digitizer estimated for the ambient runs.

17 Oct 2026 Analysis layer: everything in Julia, under `fullsim_analysis/lib/` (reading the event files, peak fits, spectra, maps, result writers, emulation on decay lists) and one script per study under `fullsim_analysis/scripts/`. The scripts write files only: `results/<campaign>/<study>/<study>.h5` for arrays, histograms and maps, CSV files for tables, no figures. Python scripts under `fullsim_analysis/plots/` read them and draw the figures beside them (JJ's caveat: plotting decoupled from analysis). Every number reported comes from a tracked script with its output under `results/`; the scratch directory is for debugging only.

17 Oct 2026 JJ: the Radioroc2 emulation comes before the overvoltage scan, on a branch `radioroc` opened from `fullsim`. Defaults taken after reading the datasheet: evaluation

board with a 12-bit ADC and the multiplexed readout; charge trigger TQ as the acquisition trigger for the slow scintillators, T1/T2 and an external majority as comparison; time-preamplifier pulse 5/20 ns as a parameter; noise 0.1 photoelectron at 100 ns falling as the inverse square root of the peaking time. The high-gain path saturates at a tenth of the low-gain one, 72 avalanches at 2.8 V, so the low-gain path is the energy measurement of the bright channels and both are written.

18 Oct 2026 Cold BGO optical run (400 000 decays, 633 s): 6421 photons at the active areas at 511 keV, FWHM 5.8% with the intrinsic term. Cold emulations: digitizer at 3 V and gain 4 with a 10 µs gate, 511 keV at 8200 photoelectrons, FWHM 6.7%, but only 4.6% of the decays triggered at 290 counts with majority 2, because the 2.7 µs decay spreads the light and the sampled amplitude of the brightest channel stays near one or two photoelectrons: the cold threshold comes from the cold trigger scan, not from the ambient one; Radioroc2 at 1.8 µs, 511 keV at 1912 avalanches, FWHM 7.2%, 1275/511 2.46. CsI(Tl) ambient with the Radioroc2: FWHM 6.0, 5.9, 5.5, 5.4% at 300, 600, 1200, 1800 ns. Analysis layer built and the first study under its rule, the dynamic range of the Radioroc2 (table in the decisions). Instructions document for the digitizer runs written.

17 Oct 2026 Tasks 0.4 to 0.12 coded and tested in one day (frozen test, tiled layout, window, materials, source and decay pipeline, photoelectron file and optical driver, SiPM model, shaping, Citiroc, digitizer, front-end stage and event file, summary and trigger-scan scripts); optical runs of both ambient configurations; first emulated campaign, table above.

17 Oct 2026 Task 0.4: frozen-output test added and passing; the digests do not depend on the thread count and change with the seed or with any parameter.

17 Oct 2026 The optical stage records every photon that reaches an active area (`pde = 1` in the kernel) and the SiPM stage applies the detection efficiency, which depends on the overvoltage and the wavelength, by thinning the records. Thinning a Poisson process is exactly equivalent to applying the efficiency photon by photon, and the overvoltage scan of task 0.13 then reuses one optical run per configuration.

17 Oct 2026 First optical run of the ambient BGO configuration on the GPU (400 000 decays, 35 min while the test suite was competing for the machine; 5000 decays take 3.7 s alone): 20.8% of the decays deposit energy, 13.0% give a 511 keV photopeak event; at 511 keV 3883 photons emitted, 3253 reach the active areas (collection 0.838), FWHM 4.1% from photon statistics alone; brightest channel median 112 photons, 1% above 451; mean map flat within 10% with the edge channels 10% lower. The acceptance of the 511 keV pair is twice that of one gamma (either partner), corrected in the design document.

17 Oct 2026 First optical run of the ambient CsI(Tl) configuration (100 000 decays, 577 s on the GPU): 15.9% of the decays deposit energy, 7.1% give a 511 keV photopeak event; 27 600 photons emitted at 511 keV, 23 200 reach the active areas (collection 0.841); brightest channel median 541 photons, 1% above 4050, which at 2.7 V (PDE 0.38 at 550 nm) is 1540 photoelectrons, above the Citiroc ceiling of 1000.

17 Oct 2026 The first front-end emulations showed that at ambient temperature the dark pulses with crosstalk companions set the trigger threshold, not the single dark photoelectron: a dark avalanche with four or more companions occurs about once per 2 µs window over the 64 channels at 3 V (crosstalk 9%) and twice at 5 V (crosstalk 19%), so a single-channel threshold of 4 to 6 photoelectrons, or 60 counts on the digitizer at 43 V, triggers on dark pulses in most windows. The threshold scan of `fullsim_analysis/trigger_scan.jl` replaces the hand estimates; the recipe for the bench must be updated accordingly.

17 Oct 2026 Task 0.5: tiled readout layout. The S14161-6050HS-04 tile is 25.0 mm (datasheet outline), so the matrix is 50.0 mm and overhangs the crystal by 1.0 mm per side, not 49.6 and 0.8 as first written; the active fraction of the footprint, 93.4%, was right. Design document corrected.